

Notes: Coles, Heath, and Ringgenberg (JFE, 2022)
On index investing

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1 Summary

Coles, Heath, and Ringgenberg study whether the rise of index investing changes information production and price efficiency. The motivating concern is straightforward: passive investors free ride on the research and trading of active investors. If too much capital becomes passive, prices might contain less firm-specific information. The paper asks whether this concern is empirically visible in stock prices.

The authors frame the question using an extension of Grossman and Stiglitz (1980). Investors can choose among three roles: passive index investors, active publicly-informed investors, and active privately-informed investors. When indexing becomes cheaper relative to active investing, more investors choose to index. The model predicts that aggregate information production falls because there are fewer active investors paying to acquire private information. However, the fraction of remaining active investors who choose to become privately informed adjusts so that price informativeness does not change.

The empirical setting uses Russell 1000/2000 index reconstitutions after Russell introduced banding in 2007. The paper develops a difference-in-differences design around the upper and lower Russell bands. Stocks that cross a band and switch indexes are compared with otherwise similar stocks that remain in their prior index. This design produces exogenous variation in index fund ownership without relying on the pre-2007 sharp regression-discontinuity design that became invalid after Russell's banding reform.

The first result is that index reconstitution materially changes investor composition. A stock switching into the Russell 2000 experiences a rise in passive fund ownership and a roughly offsetting decline in passive Russell 1000 ownership. Switching into the Russell 1000 produces the opposite pattern. This confirms that the design captures a meaningful shock to passive ownership.

The second result is that index investing changes market dynamics. Stocks that receive more index ownership experience higher turnover, greater short interest, higher return volatility, and stronger comovement with the relevant Russell index. These effects confirm that the research design has power to detect economically meaningful changes in trading and price behavior.

The third result is that index investing reduces information production. The paper studies Google search volume, SEC EDGAR page views, and buy-side analyst reports. When stocks switch into the Russell 2000 and receive more passive ownership, information production about the individual stock declines. When stocks switch out of the Russell 2000 and into the Russell 1000, the same measures increase. This result is consistent with the model's prediction that fewer active investors gather stock-specific information as indexing expands.

The fourth and central result is a non-result: price informativeness does not decline. The authors test weak-form price efficiency using variance ratios, anomaly mispricing using the Stambaugh, Yu, and Yuan style mispricing measure, and the speed of earnings-information incorporation using post-earnings-announcement drift. Across these measures, the estimated effects are statistically and economically close to zero. The authors also report minimum detectable effect sizes to show that these null estimates are informative rather than simply underpowered.

The paper's main interpretation is that passive investing changes investor composition and reduces costly information production, but active investors endogenously adjust. The remaining active investors continue to find it worthwhile to acquire information when doing so is profitable, so the link between prices and fundamentals remains stable. The evidence supports equilibrium models

in the Grossman-Stiglitz tradition rather than models predicting a first-order deterioration of price efficiency from more index investing.

2 Big picture

- **Research question.** Does an exogenous increase in index investing reduce information production and weaken the informational efficiency of stock prices?
- **Core tension.** Passive investors lower costs for end investors but free ride on active investors' information production.
- **Theory benchmark.** In an extended Grossman-Stiglitz model, more indexing reduces total information production but leaves price informativeness unchanged in equilibrium.
- **Empirical setting.** Post-2007 Russell 1000/2000 reconstitutions, where banding generates switchers and stayers around upper and lower bands.
- **Treatment.** A stock switches index membership and experiences a change in passive index fund ownership.
- **Main mechanism.** Index ownership replaces part of the active investor base, reducing the total amount of private information gathering.
- **Main conclusion.** Index investing reduces information production but does not measurably reduce price informativeness.

3 Conceptual contribution

The paper is organized around a distinction that is easy to miss. Less information production and less price informativeness are not the same thing. If the total mass of active investors falls, the economy may spend less on information acquisition. But price informativeness depends on the marginal incentives of the remaining active investors, not only on the total number of active investors.

In the authors' Grossman-Stiglitz extension, investors choose whether to be passive, active with only public information, or active with costly private information. A decline in the relative cost of indexing increases the share of passive investors. At the same time, the privately informed share among remaining active investors adjusts so that the expected return to becoming informed remains unchanged. This adjustment keeps price informativeness stable.

The model therefore generates two sharp empirical predictions:

1. An exogenous increase in index investing should reduce firm-level information production.
2. The same increase should not reduce price informativeness.

This is an important contribution because many critiques of passive investing implicitly move from "less active management" to "less efficient prices." The paper shows that the equilibrium mapping is more subtle.

4 Data and research design

4.1 Data

The paper combines Russell index membership data with CRSP, Compustat, Thomson Reuters S12 and 13F fund holdings, short interest, Google Trends search volume, SEC EDGAR page-view data, IBES analyst reports, and stock return data.

The key ownership measures separate fund ownership into passive and active components. Passive funds are identified using the CRSP index fund flag. The main passive ownership variables distinguish funds tracking the Russell 1000 from funds tracking the Russell 2000. This distinction is crucial because switching between the two Russell indexes mechanically changes the target investor clientele.

Information production is measured in three ways:

- **Google SVI:** search volume for the stock ticker.
- **EDGAR page views:** investor attention to firm filings.
- **Analyst reports:** nonduplicate buy-side analyst reports in IBES.

Price informativeness is measured in three ways:

- **Variance ratios:** deviations from a random-walk benchmark over 4- and 8-day horizons.
- **Mispricing:** the Stambaugh-style composite anomaly mispricing rank.
- **PEAD beta:** the return response to standardized unexpected earnings after earnings announcements.

4.2 Russell banding and treatment variation

Before 2007, Russell index assignment followed a sharp rank cutoff. After 2007, Russell introduced “banding” around the Russell 1000/2000 cutoff. Stocks near the band can either stay in their previous index or switch to the other index depending on their prior index membership and movement across the band.

The paper exploits this post-2007 structure. Around the lower and upper bands, treated stocks are switchers and control stocks are stayers. The authors compare switchers and stayers before and after each June reconstitution, using stock-by-cohort fixed effects and time fixed effects.

The basic difference-in-differences logic is:

$$Y_{ict} = \beta \text{IndexSwitch}_{ict} + \phi_{ic} + \lambda_t + \varepsilon_{ict},$$

where IndexSwitch_{ict} captures the stock’s switch in index assignment after reconstitution. The sign of β should reverse across the lower and upper bands because switching into the Russell 2000 increases Russell 2000 index ownership, while switching into the Russell 1000 decreases it.

5 Main findings

5.1 Finding 1: index assignment changes ownership composition

The event-time graph and regression table show that Russell reconstitution changes the composition of investors. At the lower band, stocks moving from the Russell 1000 to the Russell 2000 experience a meaningful increase in passive Russell 2000 ownership. At the upper band, stocks moving from the Russell 2000 to the Russell 1000 experience a decline in Russell 2000 passive ownership and an increase in Russell 1000 passive ownership.

This result matters because the rest of the paper relies on the idea that index assignment causes a real change in the investor base. The ownership results show that the treatment is not cosmetic.

5.2 Finding 2: index investing changes market dynamics

The paper finds that index assignment affects several market outcomes. More index ownership increases turnover, short interest, volatility, and comovement with the relevant Russell index. These effects are important for interpretation because they show that the design is powerful enough to detect market effects when they exist.

5.3 Finding 3: index investing reduces information production

The key mechanism result is in the information-production table. When stocks move into the Russell 2000 and receive more passive ownership, Google searches, EDGAR page views, and analyst reports decline. The EDGAR and analyst-report effects are statistically significant in the lower-band estimates. At the upper band, moving into the Russell 1000 produces increases in EDGAR page views and analyst reports.

The interpretation is that as index investing increases, fewer active investors produce firm-specific information. This result matches the first prediction of the model.

5.4 Finding 4: price informativeness does not decline

The central empirical result is that price informativeness is unchanged. The paper examines variance ratios, anomaly mispricing, and post-earnings-announcement drift. Across all three types of tests, index assignment has no meaningful effect.

This is not simply a weak statistical finding. The paper reports minimum detectable effect sizes and argues that the estimates are precise enough to detect economically meaningful changes. Therefore, the zero effects are informative.

5.5 Finding 5: broad trends are consistent with the local estimates

The paper ends by showing that from 2007 to 2016 passive fund ownership in Russell 3000 stocks rose sharply, while a broad variance-ratio measure did not deteriorate. The broad trend is not the identification strategy, but it is consistent with the paper's core conclusion: more passive ownership does not necessarily imply less informative prices.

6 Notes-created summary tables

Empirical structure of the paper

Component	Implementation	Purpose
Shock to investor composition	Post-2007 Russell 1000/2000 index switches around upper and lower bands	Generates variation in passive index ownership.
Treatment group	Stocks that switch indexes after re-constitution	Experiences a discrete change in index-fund clientele.
Control group	Stocks near the same band that stay in their prior index	Provides similar stocks exposed to the same local rank environment.
Fixed effects	Stock-by-cohort and calendar-time fixed effects	Remove time-invariant stock differences and aggregate time trends.
Mechanism outcomes	Google SVI, EDGAR views, analyst reports	Measure costly information production and investor attention.
Price-efficiency outcomes	Variance ratios, mispricing, PEAD beta	Test whether less information production lowers price informativeness.

Core results and interpretation

Result	Evidence	Interpretation
Investor composition changes	Passive ownership moves in the expected direction after Russell switches	The treatment generates a real passive-ownership shock.
Trading dynamics change	Turnover, short interest, volatility, and index comovement respond	The design has power and index investing affects market behavior.
Information production falls	Google searches, EDGAR views, and analyst reports decline when passive ownership rises	Fewer investors produce costly firm-specific information.
Price informativeness is unchanged	No meaningful effects on variance ratios, mispricing, or PEAD	Active investors adjust so prices remain similarly informative.
Broad trends are consistent	Passive ownership rises from 2007 to 2016 without deterioration in variance ratios	The local estimates are not contradicted by aggregate trends.

7 Interpretation and discussion

7.1 What the paper does not claim

The paper does not claim that passive investing has no market effects. It explicitly finds effects on ownership composition, turnover, short interest, volatility, correlations, and information production. The narrower claim is that these changes do not translate into a detectable decline in price informativeness in this setting.

The paper also does not prove that any possible level of passive investing is harmless. The empirical estimates are local to Russell-band switchers during 2007–2016. The authors discuss external-validity limits. Still, the evidence directly addresses a key empirical prediction: whether an exogenous passive-ownership shock lowers price informativeness for affected stocks.

7.2 Why the null result is interesting

A null effect on price informativeness is usually hard to interpret. Here it is informative because the paper first shows that the same design detects ownership and market-dynamics effects. The minimum detectable effect sizes also suggest that the tests could have found moderate price-efficiency effects if they existed.

This makes the null result theoretically meaningful. It supports the idea that information acquisition is endogenous. If passive ownership rises, some active investors leave or scale back, but the marginal incentives of remaining active investors still pin down price informativeness.

7.3 How to connect this paper to other finance research

The paper connects naturally to several literatures:

- **Grossman-Stiglitz information economics.** Prices cannot be fully revealing if information is costly; equilibrium information acquisition responds to incentives.
- **Indexing and asset demand.** Russell reconstitution affects investor clienteles and stock-level demand.
- **Market efficiency.** Price informativeness is tested through variance ratios, anomaly mispricing, and PEAD.
- **Attention and information production.** Google searches, EDGAR views, and analyst reports proxy for different dimensions of information gathering.
- **Passive ownership and governance/market outcomes.** The paper complements studies showing passive ownership affects governance, volatility, payout, or comovement.

8 Possible seminar questions

1. Is the paper’s main result best understood as “passive investing is harmless” or as “information acquisition adjusts endogenously”?

2. How should we interpret the fall in information production if price informativeness remains unchanged? Is this an efficiency gain, or could it hide welfare costs elsewhere?
3. What are the strongest external-validity concerns with Russell-band estimates?
4. Are Google searches, EDGAR views, and analyst reports measuring information production, investor attention, or both?
5. Could index assignment affect stocks through channels other than passive ownership, such as attention, liquidity, or institutional benchmarking?
6. Would the results be different for smaller stocks, less liquid stocks, or markets with fewer sophisticated arbitrageurs?
7. What would be the best way to test whether passive investing affects real investment through less informative prices?

9 Key original-paper extracts

9.1 Title, abstract, and motivation

Read: The abstract states the paper's core result: index investing lowers information production, but price informativeness remains unchanged. The introduction frames the free-riding concern: passive investors benefit from prices made informative by active managers.

Interpretation: This is a paper about an equilibrium adjustment, not a simple correlation between passive ownership and market quality.



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On index investing[☆]

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ABSTRACT

We empirically examine the effects of index investing using predictions derived from a Grossman-Stiglitz framework. An exogenous increase in index investing leads to lower information production as measured by Google searches, EDGAR views, and analyst reports, yet price informativeness remains unchanged. These findings are consistent with an equilibrium in which investors choose to gather private information whenever it is profitable. As index investing increases, there are fewer privately-informed active investors (so overall information production drops), but the mix of investors adjusts until the returns to active investing are unchanged. As a result, passive investing does not undermine price efficiency.

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Title, abstract, and article information

9.2 Model logic

Read: The model extends Grossman and Stiglitz by adding a passive index-investor choice. Investors can be passive, active-publicly-informed, or active-privately-informed.

Interpretation: The key mechanism is not that everyone stays active. Rather, the remaining active investor base adjusts until the return to private information is unchanged.

1. Introduction

Over the last two decades, the amount of capital devoted to index investing worldwide has grown by nearly 10 trillion dollars (Investment Company Institute, 2021). This increase in passive index investing is not without controversy. Passive investors are necessarily free-riding on the research and analysis performed by active managers.

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This suggests a trade-off: while passive managers allow investors to earn index returns for low fees, the effort of active managers helps ensure that prices correctly reflect fundamental value. Put differently, not everyone can index, someone has to be active. The question is, does the rise of index investing change information production in the economy? If so, does it affect the informational efficiency of stock prices?

In this paper we empirically examine the effects of index investing on information production and price efficiency. We derive predictions using a simple extension to the Grossman and Stiglitz (1980) model of endogenous information acquisition and we test these predictions using Russell Index reconstitutions as a source of exogenous variation in passive investing.¹ Over our sample period from 2007 to 2016, we find that shocks to the mix of passive and active investors cause significant changes in trading behavior and information production. Nonetheless,

¹ Wei and Young (2021) show that many existing studies that use Russell index reconstitutions suffer from selection bias. Importantly, we develop a new identification strategy that avoids these issues. We discuss this in greater detail in Section 3.

Introduction: the trade-off between low-cost indexing and information production

demand mX_{ind} shares of the asset regardless of its price – that is, indexers are price-insensitive. Thus the net supply of shares that is offered to active investors is $\bar{x} - mX_{ind}$.

The remaining mass of investors, $1-m$, are price-sensitive active investors. Each active investor chooses whether to pay an additional information cost c to learn the nonpublic signal θ . If they do, they are a privately-informed active investor and have demand for the asset:

$$X_I = \frac{1}{\psi} \frac{\theta - P}{\sigma_\epsilon^2} \quad (2)$$

If they choose not to pay c , they are a publicly-informed active investor. They have demand for the asset:

$$X_U = \frac{1}{\psi} \left(\frac{E[\tilde{\theta}|P] - P}{\text{Var}[\tilde{\theta}|P] + \sigma_\epsilon^2} \right) \quad (3)$$

The fraction of active investors that choose to pay c to become privately-informed is denoted λ . Thus, the mass of privately-informed active investors is $(1-m)\lambda$, and their total expenditure on information production is $(1-m)\lambda c$.

In equilibrium, each investor is indifferent between the three categories:

$$\begin{aligned} E[-\exp(-\psi \tilde{W}_1) | X_{index}] &= E[-\exp(-\psi \tilde{W}_1) | \tilde{X}_I] \\ &= E[-\exp(-\psi \tilde{W}_1) | \tilde{X}_U] \end{aligned} \quad (4)$$

That is, the expected CARA utility of final wealth for an index investor, a publicly-informed active investor, and a privately-informed active investor is the same. We solve for the equilibrium numerically (details are in the Internet Appendix).

As in [Grossman and Stiglitz \(1980\)](#), we find that the correlation ρ between price and fundamental value

Model equations and equilibrium indifference

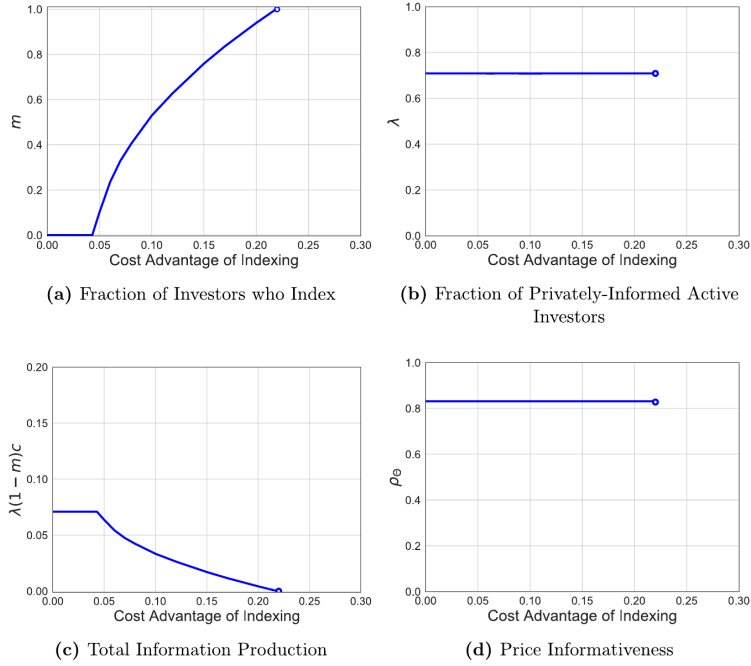
the informativeness of the price is determined by the *ratio* of active privately-informed investors, who correct mispricing, to active publicly-informed investors who generate mispricing. While an increase in the quantity of index investors leads to less active investing overall, as shown in Panel B, the ratio of active privately-informed to active publicly-informed investors is constant for any level of index investing. As a result, the quantity of index investors does not matter because the remaining investors will adjust their behavior until price informativeness is unchanged.

Remark 4. An exogenous increase in index investing leads to no change in price informativeness.

[Remark 4](#) motivates our empirical analyses on price informativeness: following an increase in index investors, we expect to see no change in price informativeness. Again, the intuition is simple: investors will always choose to become privately-informed active traders if it is profitable to do so. Importantly, this result holds for any size change in the quantity of index investors (i.e., for small or large increases in index investing we expect to see the same thing – no change in price informativeness).

Of course, our model is necessarily a simplified depiction of the market. In reality, funds might not perfectly fit into one of our three investor categories: passive, publicly-informed active, and privately-informed active. For exam-

Remark 4: more indexing does not change price informativeness



1. Model Predictions. The figure plots comparative statics of the market equilibria in our version of the [Grossman and Stiglitz \(1980\)](#) model. The x-axis shows changes in the market equilibrium to the cost advantage of index investing relative to active investing ($c_i - c_a$) starting from zero and increasing.

Figure 1: model predictions for indexing, private information, information production, and price informativeness

9.3 Data and sample construction

Read: The paper combines Russell index data with fund holdings, CRSP/Compustat, short interest, Google search, EDGAR views, and analyst reports. Table 1 shows that upper- and lower-band firms are narrow slices of mid-cap stocks.

Interpretation: The identification depends on comparing local switchers and stayers, not the full population of Russell firms.

firm data from Compustat, ownership data from Thomson Reuters S12 and 13F, and Russell index data directly from Russell Investments.

3.1. Data

Russell index membership data come directly from Russell. Financial data on our sample of stocks are from CRSP and the merged CRSP-Compustat database.

From Compustat, we also obtain information on firm-level short interest. Short interest data are compiled by the stock exchanges twice per month, near the middle and end of each month, and the data are publicly released four business days later. Prior to September of 2007, short interest data were only reported once per month, near the middle of the month, so for consistency we use mid-month short interest in all of our analyses.¹⁶ We also use information on anomaly mispricing using the mispricing score from [Stambaugh et al. \(2015\)](#), available on Robert Stambaugh's website. We obtain daily search volume for each stock ticker from Google Trends. We obtain data on analyst coverage from IBES, and data on EDGAR search volume from the SEC EDGAR log files.

We classify funds as active or passive using the CRSP index fund flag (index fund flag = "D"). This definition includes open-ended mutual funds, closed-end funds and ETFs (which we refer to collectively as "funds"); a fund is designated as an index fund if its objective is to replicate an index and not to generate abnormal returns. Neither our model nor our tests make any assumptions about the form of passive investing (i.e., index funds can be mutual funds, closed-end funds, or ETFs).

Fund holdings data come from the Thomson Reuters S12 and CRSP databases. We compute the ownership of each sample stock by every fund in each quarter, using the union of the two datasets: When one of the two holdings datasets records fund X as holding N shares of firm Y in

Table 1
Summary Statistics.
The table presents summary statistics for ex ante stock characteristics: market capitalization, monthly trading turnover, monthly volatility, and categories of fund ownership. The sample consists of stocks that were potential switchers within a +/-100 rank window of the yearly Russell upper and lower bands from 2007 to 2016.

Panel A: Lower Band						
	Mean	Std. Dev.	p10	p50	p90	N
Market Cap (\$M)	1630.31	499.43	902.45	1647.34	2295.67	319
FundOwn ^{active}	0.07	4.51	2.93	9.27	14.55	319
FundOwn ^{passive}	22.71	12.12	6.81	22.86	38.90	319
FundOwn	31.78	14.13	12.89	32.07	50.05	319
logTurn	-1.33	0.85	-2.29	-1.35	-0.13	319
LogVt	-3.79	0.6	-4.49	-3.88	-2.97	319
ShortInt	0.09	0.08	0.02	0.07	0.21	319
ρ_{1000}	0.47	0.24	0.12	0.51	0.77	319
ρ_{2000}	0.47	0.24	0.12	0.52	0.75	319
MarketBeta	1.38	0.88	0.45	1.26	2.48	319
logSVI	3.49	0.91	2.56	3.71	4.36	319
logEDGAR	5.42	1.55	3.82	5.76	6.85	319
logAnalystReports	2.18	0.80	1.10	2.30	3.00	319
Panel B: Upper Band						
	Mean	Std. Dev.	p10	p50	p90	N
Market Cap (\$M)	3019.27	875.14	1894.60	2946.14	4153.87	564
FundOwn ^{active}	9.12	4.63	3.37	8.06	15.54	564
FundOwn ^{passive}	24.50	11.73	8.11	24.73	39.52	564
FundOwn	33.62	13.74	14.43	33.91	51.81	564
logTurn	-1.48	0.73	-2.33	-1.46	-0.66	564
LogVt	-3.91	0.50	-4.48	-3.94	-3.27	564
ShortInt	0.07	0.05	0.02	0.06	0.14	558
ρ_{1000}	0.52	0.23	0.23	0.56	0.78	564
ρ_{2000}	0.55	0.23	0.23	0.61	0.81	564
MarketBeta	1.37	0.75	0.52	1.32	2.36	564
logSVI	3.46	0.86	2.30	3.71	4.29	564
logEDGAR	5.07	1.75	3.53	5.51	6.69	564
logAnalystReports	2.09	0.75	1.10	2.30	2.83	564

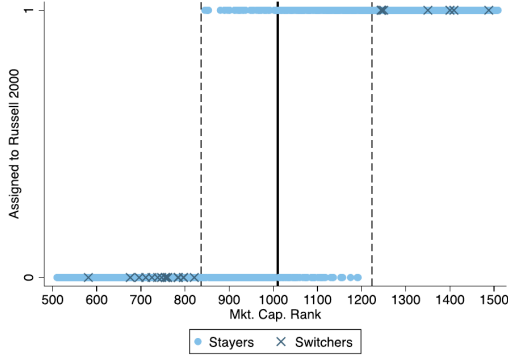
Table 1: summary statistics for lower- and upper-band samples

Data sources and ownership variable construction

9.4 Russell assignment and the treatment-control comparison

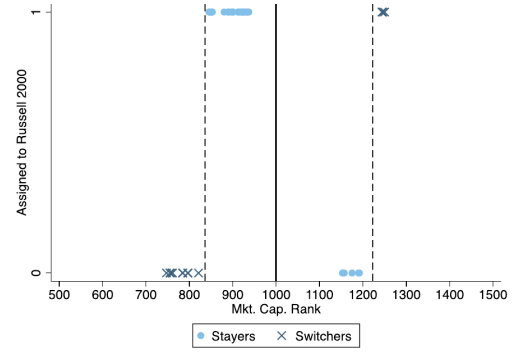
Read: Figures 2 and 3 show how banding creates switchers and stayers around the Russell 1000/2000 bands. The design avoids relying on the old sharp discontinuity.

Interpretation: The visual logic is central: near the bands, some stocks switch indexes and some do not, even though they are close in rank.



Index Assignments. The figure plots stocks' assignments to the Russell 1000 and 2000 indexes in June of 2007 against market cap rankings. Stocks near the rank-1000 threshold (solid line) all stayed in their previous index, breaking close to the estimated upper and lower bands (dashed lines), however, there are sharp discontinuities in index switch

Figure 2: 2007 Russell index assignment under banding



struction. The figure plots the sample stocks' assignment to the Russell 1000 and 2000 indexes in June of 2007 against market cap rankings. The sample consists of stocks that were potential switchers within a +/-100 rank window of the yearly

Figure 3: treated switchers and control stayers in the sample

Russell's proprietary market cap rankings. The sample consists of stocks in switch indexes from the previous year based on their lagged index members

year before and one year after assignment. We estimate difference-in-differences specifications, with fixed effects for each stock in each cohort. For the upper-band treatment effect, the specification is:

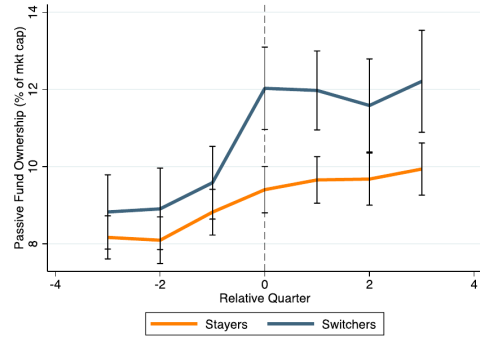
$$Y_{ict} = \beta \times I\{R2000 \rightarrow R1000\}_{ic} \times PostAssignment_{ct} + \phi_{ic} + \lambda_t + \epsilon_{ict}, \quad (6)$$

and for the lower-band treatment effect the specification is:

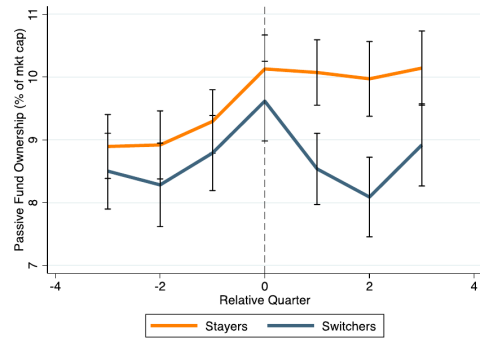
$$Y_{ict} = \beta \times I\{R1000 \rightarrow R2000\}_{ic} \times PostAssignment_{ct} + \phi_{ic} + \lambda_t + \epsilon_{ict}, \quad (7)$$

where ϕ_{ic} and λ_t are stock-by-cohort and time fixed effects. The firm-by-cohort fixed effects sweep out any time-invariant differences between treated and control stocks. Importantly, this includes preexisting differences on any characteristics between treated and control stocks. For ex-

Difference-in-differences equations for upper and lower bands



(a) Lower Band



(b) Upper Band

ownership in Event Time. The figure plots the average ownership by passive funds of control (stayer) and treated (switcher) stocks relative to index assignment. The figure also shows 95% confidence intervals for each quarterly mean. The sample consists of stocks that were potential switchers within a +/-100 rank window of the yearly

Figure 4: event-time change in passive ownership

9.5 Fund ownership and market dynamics

Read: Table 2 verifies that index switches shift passive ownership in the expected direction. Table 3 shows effects on turnover, short interest, volatility, index correlations, and market beta.

Interpretation: These results establish that the treatment meaningfully changes investor composition and market dynamics before the paper turns to information and price efficiency.

Table 2
Difference-in-differences Regression of Fund Ownership around Changes in Index Assignment.
The table presents difference-in-differences estimates of the effects of Russell index switching on fund ownership, expressed as a percentage (1–15) of the stock's market capitalization. The estimating equation is:

$$FundOwn_{it} = \beta IndexSwitch_{it} + \phi_t + \lambda_i + \epsilon_{it},$$

where $IndexSwitch_{it}$ equals one for stocks that switched indexes in quarters after the June Russell index reconstitution, and zero otherwise. $FundOwn_{it}^{2000}$, $FundOwn_{it}^{1000}$ denote ownership by passive funds that track the Russell 2000 and 1000 respectively. $FundOwn_{it}^{passive}$ is total ownership by passive funds. $FundOwn_{it}^{active}$ is total ownership by active funds. $FundOwn_{it}$ is total ownership by all funds. $InstOwn_{it}$ is total ownership by institutional investors that filed a 13F statement. The sample consists of quarterly observations for 3 quarters pre- and post- index reconstitution, for stocks that were potential switchers within a +/-100 rank window of the two Russell bands, each year from 2007 to 2016. Standard errors are robust and clustered by stock. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

Panel A: Lower Band						
	(1) $FundOwn_{it}^{2000}$	(2) $FundOwn_{it}^{1000}$	(3) $FundOwn_{it}^{passive}$	(4) $FundOwn_{it}^{active}$	(5) $FundOwn_{it}$	(6) $InstOwn_{it}$
$R1000 \rightarrow R2000_{it} \times$ $PostAssignment_{it}$	1.81*** (26.5)	-0.23*** (-24.2)	1.81*** (7.4)	-0.45 (-0.6)	1.36 (1.6)	5.33 (1.2)
Observations	1914	1914	1914	1914	1914	1914
Adjusted R-squared	0.801	0.820	0.908	0.851	0.865	0.739
Stock $\times$ Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Upper Band						
	(1) $FundOwn_{it}^{2000}$	(2) $FundOwn_{it}^{1000}$	(3) $FundOwn_{it}^{passive}$	(4) $FundOwn_{it}^{active}$	(5) $FundOwn_{it}$	(6) $InstOwn_{it}$
$R2000 \rightarrow R1000_{it} \times$ $PostAssignment_{it}$	-1.63*** (-28.0)	0.23*** (34.8)	-1.16*** (-5.5)	1.18* (1.8)	0.01 (0.0)	-3.76*** (-3.2)
Observations	3384	3384	3384	3384	3384	3384
Adjusted R-squared	0.817	0.844	0.880	0.837	0.833	0.863
Stock $\times$ Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year $\times$ Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes

Table 2: effects of index assignment on fund ownership

Table 3
Effects of Index Assignment on Asset Markets.
The table presents difference-in-differences estimates of the effects of index assignment on the markets for individual stocks. The estimating equation is:

$$Y_{it} = \beta IndexSwitch_{it} + \phi_t + \lambda_i + \epsilon_{it},$$

where $IndexSwitch_{it}$ equals one for stocks that switched indexes in months after the June Russell index reconstitution, and zero otherwise. $LogTurn_{it}$ is the log of monthly volume divided by shares outstanding. $ShortInt_{it}$ is the mid-month short interest divided by shares outstanding. $LogVt$ is the log of daily return volatility. ρ^{1000} and ρ^{2000} are the daily return correlation with the Russell 1000 and 2000 index respectively. $MarketBeta_{it}$ is the beta of the stock's daily returns on the daily CRSP value-weighted market return. MDES and Sample St.Dev. are the minimum detectable effect size and the sample standard deviation of each outcome variable. The sample consists of monthly observations for 11 months before and after index assignment, for stocks that were potential switchers within a +/-100 rank window of the two Russell bands each year from 2007 to 2016. Standard errors are robust and clustered by stock. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

Panel A: Lower Band						
	(1) $LogTurn_{it}$	(2) $ShortInt_{it}$	(3) $LogVt_{it}$	(4) ρ^{1000}	(5) ρ^{2000}	(6) $MarketBeta_{it}$
$R1000 \rightarrow R2000_{it} \times$ $PostAssignment_{it}$	0.121*** (2.8)	0.019*** (3.2)	0.064* (1.7)	0.046*** (2.9)	0.069*** (4.5)	0.300*** (4.5)
MDES	±0.122	±0.017	±0.102	±0.044	±0.043	±0.186
Sample St.Dev.	0.828	0.081	0.612	0.237	0.233	0.986
Observations	6643	6614	6642	6450	6450	6642
Adjusted R-squared	0.819	0.807	0.687	0.394	0.388	0.278
Stock × Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Upper Band						
	(1) $LogTurn_{it}$	(2) $ShortInt_{it}$	(3) $LogVt_{it}$	(4) ρ^{1000}	(5) ρ^{2000}	(6) $MarketBeta_{it}$
$R2000 \rightarrow R1000_{it} \times$ $PostAssignment_{it}$	-0.003*** (-2.9)	-0.015*** (-4.6)	-0.049** (-2.1)	-0.035*** (-3.9)	-0.053*** (-5.7)	-0.175*** (-4.5)
MDES	±0.088	±0.009	±0.065	±0.025	±0.026	±0.107
Sample St.Dev.	0.725	0.056	0.568	0.226	0.224	0.748
Observations	11,670	11,530	11,669	11,302	11,302	11,669
Adjusted R-squared	0.791	0.784	0.665	0.398	0.390	0.289
Stock × Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Year × Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Table 3: effects of index assignment on asset markets

9.6 Information production

Read: Table 4 shows the effect of index assignment on Google SVI, EDGAR page views, and analyst reports.

Interpretation: The table supports the mechanism: when passive ownership rises, stock-specific information production falls.

Table 4
Effects of Index Assignment on Information Production.
The table presents difference-in-differences estimates of the effects of index assignment on measures of information production about individual stocks. The estimating equation is:

$$Y_{it} = \beta IndexSwitch_{it} + \phi_t + \lambda_i + \epsilon_{it},$$

where $IndexSwitch_{it}$ equals one for stocks that switched indexes in months (quarters) after the June Russell index reconstitution, and zero otherwise. $log(SVI)_{it}$ is the log of one plus the monthly average Google Search Volume Index for the stock's ticker. $log(EDGAR)_{it}$ is the log of one plus the monthly average number of page requests per day for the firm's filings on the SEC EDGAR website. $log(AnalystReports)_{it}$ is the log of one plus the quarterly number of nonduplicate equity analyst reports about the firm recorded in I/B/E/S. MDES and Sample St.Dev. are the minimum detectable effect size and the sample standard deviation of each outcome variable. For the first two measures the sample consists of monthly observations for 11 months before and after index assignment, for stocks that were potential switchers within a +/-100 rank window of the two Russell bands each year from 2007 to 2016. For the third measure the sample consists of quarterly observations for 3 quarters before and after index assignment. Standard errors are robust and clustered by stock. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

Panel A: Lower Band			
	(1) $log(SVI)_{it}$	(2) $log(EDGAR)_{it}$	(3) $log(AnalystReports)_{it}$
$R1000 \rightarrow R2000_{it} \times$ $PostAssignment_{it}$	-0.038 (-1.0)	-0.246* (-1.9)	-0.107* (-1.8)
MDES	± 0.104	± 0.366	± 0.164
Sample St.Dev.	0.822	1.877	0.853
Observations	6699	6699	1914
Adjusted R-squared	0.836	0.787	0.717
Stock $\times$ Cohort FE	Yes	Yes	Yes
Year $\times$ Month FE	Yes	Yes	No
Year $\times$ Quarter FE	No	No	Yes
Panel B: Upper Band			
	(1) $log(SVI)_{it}$	(2) $log(EDGAR)_{it}$	(3) $log(AnalystReports)_{it}$
$R2000 \rightarrow R1000_{it} \times$ $PostAssignment_{it}$	0.043 (1.4)	0.200*** (2.5)	0.143*** (2.8)
MDES	± 0.083	± 0.224	± 0.140
Sample St.Dev.	0.897	1.957	0.825
Observations	11,844	11,844	3384
Adjusted R-squared	0.830	0.809	0.598
Stock $\times$ Cohort FE	Yes	Yes	Yes

Table 4: effects of index assignment on information production

9.7 Price informativeness

Read: The paper measures price informativeness using variance ratios, mispricing, and PEAD. Table 5 shows no meaningful effect of index assignment on these measures.

Interpretation: This is the key contrast in the paper: less information production does not imply less informative prices in equilibrium.

4.6. Effects on price informativeness

A number of theoretical models examine the relation between index investing and price efficiency. Notably, [Remark 4](#) of our model predicts a non-result: in equilibrium, the fraction of active investors who pay to become privately-informed should adjust to a change in the composition of investors such that price informativeness is unchanged. Accordingly, we next examine the effects of passive investing on the information content of stock prices. First, we examine weak-form price efficiency. We compute variance ratios for each month of sample stocks' returns over horizons of $q = 4$ and 8 trading days. Formally, we use the q -period bias-corrected variance ratio test of [Lo and MacKinlay \(1988\)](#):

$$\text{VarRatio}(q) = \frac{\hat{\sigma}^2(q)}{q \times \hat{\sigma}^2}, \quad (8)$$

where

$$\hat{\sigma}^2(q) = \frac{k}{(n - q + 1)(k - 1)} \sum_{t=q}^n (p_t - p_{t-q} - q\hat{\mu})^2, \quad (9)$$

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_{t=1}^n (p_t - p_{t-1} - \hat{\mu})^2, \quad (10)$$

$$\hat{\mu} = \frac{1}{n} \sum_{t=1}^n (p_t - p_{t-1}), \quad (11)$$

and the data consists of $kq + 1$ observations (for convenience we define $n = kq$). We calculate variance ratios separately for each firm and year using overlapping observations within the month. Formally, we examine the absolute value of the centered ratio:

$$\text{AbsVarRatio}_t(q) = \text{abs}(\text{VarRatio}_t(q) - 1) \quad (12)$$

The efficient benchmark is a variance ratio equal to 1

Variance-ratio definition and weak-form price efficiency test

estimated effect is less than one-twentieth of one standard deviation. Moreover, the minimum detectable effect size (MDES) for the AVR in this estimate is ± 0.068 , so our research design has the ability to reliably detect a treatment effect on the order of one-sixth of one standard deviation. Yet we find no effect – precisely estimated. The results are similar for both the 8-day AVR and in the upper-band subsample.

Next we investigate stock-level mispricing. To do this, we use the mispricing measure of [Stambaugh et al. \(2015\)](#), which measures the combined ranking of each stock on 11 documented asset-pricing anomalies each month. The average mispricing level (*misp*) for our sample of stocks is 0.501 compared to a level of 0.503 across the entire universe of CRSP stocks used by [Stambaugh et al. \(2015\)](#). Thus, our sample of mid-cap stocks near the Russell thresholds appears to be representative of the broader universe of stocks in terms of mispricing.

[Table 5](#) Column 3 shows that in both subsamples, a change in passive investing in the stock was not associated with any significant change in the anomaly-mispricing score of the stock. Again, we note that the insignificant effects on *Misprice* are unlikely a result of insufficient power. We see from the MDES that our research design has enough statistical power to detect a treatment effect on the order of one-third of one standard deviation in the lower-band sample, and one-fifth of one standard deviation in the upper-band sample – five percentile ranks and three percentile ranks on a range of 1–100, respectively. Yet in both cases we find no effect. Thus, using the mispricing measure of [Stambaugh et al. \(2015\)](#) we again find a zero treatment effect that is precisely estimated, indicating that index investment does not significantly change treated stocks' price efficiency.

Finally, we investigate the incorporation of information into stock prices. To do this, we use a simple and well-documented source of return predictability that varies

Mispricing and PEAD-based price-informativeness tests

Table 5
Effects of Index Assignment on Price Informativeness.
The table presents difference-in-differences estimates of the effects of index assignment on measures of price informativeness. The estimating equation is:

$$Y_{it} = \beta \text{IndexSwitch}_{it} + \phi_i + \lambda_t + \epsilon_{it},$$

where IndexSwitch_{it} equals one for stocks that switched indexes in months (quarters) after the June Russell index reconstitution, and zero otherwise. AbsVarRatio_t^4 and AbsVarRatio_t^8 are the 4-day and 8-day absolute variance ratios of price changes. Misprice_t is the mispricing measure of [Stambaugh et al. \(2015\)](#). β^{PEAD} is the coefficient of the stock return over the three days around earnings announcements, regressed on the standardized unexpected earnings (SUE). MDES and Sample St.Dev. are the minimum detectable effect size and the sample standard deviation of each outcome variable. The sample consists of monthly observations for 11 months (PEAD: 24 months) before and after index assignment, for stocks that were potential switchers within a ± 100 rank window of the two Russell bands each year from 2007 to 2016. Standard errors are robust and clustered by stock. ***: $p < 0.01$, **: $p < 0.05$, *: $p < 0.1$.

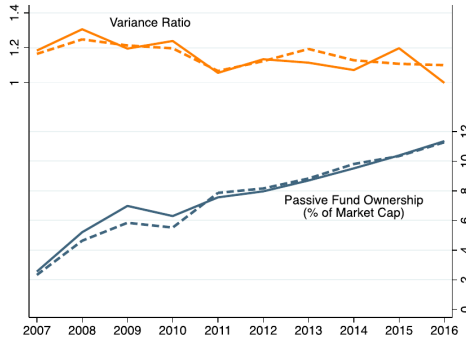
Panel A: Lower Band				
	(1) AbsVarRatio_t^4	(2) AbsVarRatio_t^8	(3) Misprice_t	(4) β_t^{PEAD}
$R1000 \rightarrow R2000_t \times$	-0.018	-0.023	-0.000	-0.034
$\text{PostAssignment}_{it}$	(-0.8)	(-0.3)	(-0.0)	(-1.0)
MDES	± 0.068	± 0.222	± 0.046	± 0.096
Sample St.Dev.	0.416	1.367	0.137	0.102
Observations	6482	5238	434	434
Adjusted R-squared	0.068	0.102	0.737	0.232
Stock $\times$ Cohort FE	Yes	Yes	Yes	Yes
Year $\times$ Month FE	Yes	Yes	Yes	No
Year FE	No	No	No	Yes
Panel B: Upper Band				
	(1) AbsVarRatio_t^4	(2) AbsVarRatio_t^8	(3) Misprice_t	(4) β_t^{PEAD}
$R2000 \rightarrow R1000_t \times$	0.012	0.008	0.010	-0.014
$\text{PostAssignment}_{it}$	(0.8)	(0.2)	(1.2)	(-0.5)
MDES	± 0.041	± 0.133	± 0.025	± 0.077
Sample St.Dev.	0.387	1.260	0.127	0.185
Observations	11,364	11,363	9452	845
Adjusted R-squared	0.064	0.095	0.742	0.222
Stock $\times$ Cohort FE	Yes	Yes	Yes	Yes
Year $\times$ Month FE	Yes	Yes	Yes	No

Table 5: effects of index assignment on price informativeness

9.8 Overall trends and conclusion

Read: Figure 5 plots rising passive ownership alongside variance-ratio trends. The conclusion states that index investing affects market dynamics and lowers information production but does not reduce price informativeness.

Interpretation: The broad trend is consistent with the local Russell evidence, though the causal evidence comes from the difference-in-differences design.



Is in Passive Ownership and Price Efficiency. The figure plots the average 4-day return variance ratio and average p

Figure 5: passive ownership rises while variance ratios do not deteriorate

stock's market capitalization. Dashed lines indicate the equal-weighted average; solid lines indicate the value-weighted average.

ment drift, or anomaly mispricing. The results confirm an important, but untested, prediction from [Grossman and Stiglitz \(1980\)](#): Following an exogenous change in investor composition, we find evidence that the mass of active informed investors adjusts such that price informativeness is unchanged.

We are careful to note that both our theoretical and empirical results rely on assumptions. Theoretically, we assume investors can be categorized into one of three groups. In practice, investors may not fall perfectly into one category (e.g., [Robertson, 2019](#)). Moreover, as noted by [Pedersen \(2018\)](#), active funds do not have to underperform passive funds after fees because real-world rebalancing frictions can affect the performance of both passive and active funds.

Empirically, we caution that the external validity of our results remains unknown. The difference-in-differences methodology we employ estimates the local average treatment effect (LATE) of index assignment around the inner

the model's sharpest prediction is that there is no effect on price informativeness, which is what we find in the data.

Thus, notwithstanding these caveats, we believe our estimates provide important data about an important question. More broadly, we note that our model's theoretical predictions hold for any feasible level of passive investing, which suggests our estimates might hold for larger or smaller changes in index investing. In addition, the overall trends in index fund ownership and price efficiency in recent years are consistent with our predictions and our cleanly identified estimates. Fig. 5 plots the yearly average levels of passive ownership and inverse price efficiency (measured by the 4-day variance ratio) across all Russell 3000 stocks during our sample period from 2007 to 2016. In this ten year period, index fund ownership quintupled, from 2% to 11% of market capitalization on average. Yet over the same period, the average variance ratio fell slightly from 1.17 to 1.10, reflecting a slight increase

Conclusion: less information production, no decline in price informativeness

10 What to remember

- The paper distinguishes **information production** from **price informativeness**.
- The model predicts that more indexing reduces the total amount of costly information gathering, but does not reduce price informativeness because active investors adjust.

- The empirical design uses post-2007 Russell 1000/2000 banding, not the old sharp cutoff RD.
- Index switches materially change passive ownership and market dynamics, so the design has empirical power.
- Information production falls when index ownership rises, measured by Google searches, EDGAR views, and analyst reports.
- Price informativeness does not fall, measured by variance ratios, anomaly mispricing, and PEAD.
- The main contribution is an equilibrium interpretation of passive investing: fewer active investors can still be enough if the marginal return to becoming informed remains unchanged.